

Jargon Buster

3D textures: An integral part of the nFiniteFX Engine, it makes hollow objects solid with true 3D material properties such as wood grain or marbling.

Accuvview: An advanced multisampling solution, which provides wider data paths to handle extra virtual pixels without slowing down rendering speeds.

Advanced Pixel Shaders: A part of the nFiniteFX II Engine that produces realistic lighting and surface effects.

Anti-aliasing: Reduces 'jaggies' at the edges of textures in a 3D game. Modern cards support up to 4x AA.

Bump mapping: A shading technique that uses multiple textures and lighting effects to simulate wrinkled or bumped surfaces so that effects such as ripples in a pond appear true to life.

Frame buffer: The part of video memory which stores pixel information of one complete screen data.

Memory pre-charge: A technique by means of which a GPU determines what part of its memory needs to be accessed next and have that section ready to accept a write or read function.

NSR: Short for nVidia Shading Rasterizer, it imparts realistic material properties like wrinkles on a model's face, fur etc through per-pixel shading.

Occlusion culling: A technique to improve performance by not rendering the objects in a 3D scene that are hidden from the human eye.

Programmable shader: These can be configured with software commands and can be used to create an infinite variety of special effects.

Quincunx: A new anti-aliasing (AA) technique that bridges the gap between the great quality of 4x AA and the performance of 2x AA.

RAMDAC: Short for Random Access Memory Digital to Analog Converter, it converts the digital data stored in the frame buffer into analog data that can be sent directly to an analog monitor.

Vertex shaders: This is one half of nVidia's nFiniteFX Engine that breathes life into characters in a 3D environment by exhibiting realistic effects such as wrinkles on a human face.

Z-Buffer: The area of the graphics memory, which stores the Z or depth information about rendered objects and foreground objects in the frame buffer.

nFiniteFX II Engine, which is a refinement of the one we saw in its predecessor. It has dual vertex shaders (one more than the GeForce3) and advanced pixel shaders that use a new technique called Z-correct bump to generate incredibly detailed surfaces. The GeForce4 core also features four independent pixel pipelines, each of which is capable of processing two textures per pixel per cycle.

A significant improvement over the GeForce3 is the ability to run any game with anti-aliasing 'ON', without a significant performance hit. The ultimate goal is to make anti-aliasing—which eliminates jagged edges in curved surfaces—an integral part of the gaming experience. This is now achieved by Accuvview, which is the GeForce's anti-aliasing system. It combines a new sampling technique that improves quality significantly with a filtering technique that reduces the performance hit to a great extent.

The wonder of it all

The biggest improvement over the



Check out the propeller wash thrown by the 'copter in Comanche 4. With a GeForce4 it's as real as in the movies!

GeForce3, however, is in the way the GeForce4 uses its onboard memory. If a video card isn't being fed with data at all times even the fastest graphics processor can do little to improve performance. LMA (Light Memory Architecture), which has been present from the GeForce3 days along with its crossbar memory controller, managed to save up on a lot of wasted bandwidth. LMA II takes that ideology a step further by using a combination of really clever tricks to eliminate memory bandwidth bottlenecks. There is a crossbar-based memory controller that uses a form of load balancing to ensure that all memory requests by the graphics processor are handled properly. Under complex loads, LMA II's memory crossbar architecture delivers between two and four times the memory bandwidth of other standard architectures.

Then there's the Quad Cache memory caching subsystem, which consists of high-speed access buffers that store small amounts of data and operate at tremendously high bandwidth. They make it possible for data to be queued in the right order so

